

**Development of Predictive Economic Decision-Support
Frameworks for Modernizing U.S. Trade and Industrial
Policy and Strengthening Domestic Manufacturing
Resilience and Competitiveness**

A Research Paper

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ABSTRACT

The U.S. has extensive federal data on trade, manufacturing, and economic activity, but these resources remain fragmented across agencies and often lack sufficient industry-level detail, timely indicators, and information on lower-tier supply chains. Most available data are also designed to describe past conditions rather than predict future risks. These limitations make it difficult for policymakers, economic-development organizations, and manufacturers to anticipate how trade restrictions, foreign supply disruptions, and other shocks may affect connected U.S. industries. This study develops a Predictive Trade and Industrial Intelligence Framework that combines trade, manufacturing, supply-chain, investment, labor, and energy data with econometric and machine-learning forecasting, supply-chain risk analysis, and policy-scenario assessment.

The framework was tested using a real panel of 19 U.S. manufacturing subsectors covering 2015 to 2025, with 209 industry-year observations from BEA, BLS, and USITC data. The two-way fixed-effects model showed a positive association between trade activity and output, but this was not statistically significant under the wild-cluster bootstrap. Import-origin concentration was generally negatively associated with output, although it was not significant in the preferred specification or in the trade-by-concentration model. The forecasting exercise also found that the more complex models were difficult to improve upon the naive persistence benchmark.

The main contribution of the study is the integration of fragmented federal data into a single analytical panel alongside econometric modeling, forecasting evaluation, and a decision-support prototype. The findings provide preliminary evidence for the first three phases of the proposed five-phase framework, while also identifying areas requiring further development, including formal supply-chain network mapping, causal scenario analysis, and broader retrospective validation.

Keywords: trade policy, industrial resilience, supply-chain risk, predictive analytics, econometric forecasting, scenario simulation, manufacturing competitiveness, decision-support systems.

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CHAPTER ONE: INTRODUCTION

1.1 Background and Context

The global network of suppliers, manufacturers, distributors, and end markets in which the United States (U.S.) economy participates has grown increasingly important to the country's economic security. Supply chains that are efficient in stable environments but brittle under stress can result from trade liberalization, outsourcing, and the geographic concentration of critical manufacturing capacity. An overseas manufacturing closure, an export restriction on a critical mineral, a transportation bottleneck, or a geopolitical crisis can quickly propagate through supplier tiers and affect domestic output, employment, and prices. Recent disruptions have repeatedly illustrated these dependencies: shortages of pharmaceutical ingredients exposed vulnerabilities in health-related supply chains; critical-mineral and rare-earth restrictions highlighted dependencies in clean-energy, defense, and advanced-manufacturing industries; and semiconductor disruptions constrained automobile and electronics production. Each episode underscored the same challenge: policymakers often lack sufficiently detailed, forward-looking, and integrated supply-chain intelligence to anticipate disruptions rather than merely respond to them.

The federal statistical infrastructure—including the U.S. Census Bureau, Bureau of Economic Analysis (BEA), Bureau of Labor Statistics (BLS), U.S. International Trade Commission (USITC), Office of the U.S. Trade Representative (USTR), Department of Commerce, and Federal Reserve—produces a vast volume of trade, manufacturing, investment, labor, and macroeconomic data that constitutes a major national asset. However, these systems were largely designed for retrospective description rather than predictive, supply-chain-aware decision support. Data remain dispersed across agencies, are released on different schedules, are often aggregated above the level needed to trace lower-tier relationships, and are not routinely integrated into models designed to assess how shocks may propagate across industries.

This paper fills that void. In order to integrate disparate federal and industry data, apply econometric and predictive modeling, simulate policy and disruption scenarios, and translate results into decision-support tools for policymakers, economic development agencies, and manufacturers, it creates and documents a Predictive Trade and Industrial Intelligence Framework. This framework is intended to be nationwide in scope, modular across industries and regions, and updated on a regular basis.

1.2 Statement of the Problem

The usefulness of federal data for anticipatory decision-making is diminished by a number of structural limitations: information is dispersed across agencies that are not intended to work together; published statistics are frequently too coarse to reveal specific inputs, suppliers, or components; many indicators are lagging by design; lower-tier supplier visibility, the second-, third-, and fourth-tier firms where disruptions frequently originate is limited; and current tools typically describe past conditions rather than forecast how a prospective shock would spread.

As a result, trade policy, resilience investment, and competitiveness strategy are often developed using a retroactive, partial view. Manufacturers might not be able to measure supplier concentration, policymakers might not be able to predict vulnerability to a potential disruption, and economic development organizations could not have enough data to allocate investment where it matters most.

For the stakeholders in charge of U.S. trade policy, manufacturing resilience, and industrial competitiveness, there is a documented, unfulfilled need for a scalable analytical framework that integrates various economic data, applies predictive and scenario-based modeling, and translates results into actionable insight.

1.3 Purpose of the Study

With a replicable methodology that can be expanded beyond the initial pilot scope, the goal of this study is to design, document, and pilot-test a predictive economic decision-support framework that can integrate trade, manufacturing, supply-chain, labor, and energy data; forecast trade flows, production, and manufacturing conditions; identify supply-chain concentration and dependency risk; simulate the economic effects of disruptions and policy interventions; and present results through tools that manufacturers, policymakers, and economic development organizations can use.

1.4 Research Objectives

This study pursues the following objectives:

- i. To review the existing state of U.S. commerce and identify specific deficiencies in granularity, timeliness, interoperability, and lower-tier supply-chain visibility.
- ii. To map the relationships among overseas suppliers, imported inputs, domestic manufacturers, and downstream industries in order to identify crucial dependencies and potential bottlenecks.

- iii. To build econometric, time-series, and machine-learning models capable of forecasting trade flows, domestic output, investment, import dependency, and manufacturing conditions.
- iv. To create supply-chain risk models that can measure domestic capacity gaps, supplier concentration, and critical-input dependency.
- v. To create scenario-simulation and stress-testing models that can assess the impact of different trade and industrial-policy interventions, tariffs, international export limits, shortages, demand shocks, and geopolitical disruptions.
- vi. To create and verify decision-support dashboards and indicators that convert model output into a format that manufacturers, policymakers, and economic development organizations may use.
- vii. To provide a modular, replicable technique that enables the framework to be expanded to other sectors and geographical areas.

1.5 Research Questions

The following research questions serve as the framework for the study:

RQ1: What are the main limitations in the current government and industry data infrastructure that prevents supply-chain-aware, predictive analysis of U.S. manufacturing and trade conditions?

RQ2: How can data on labor, energy, manufacturing, supply-chain, trade, and investment be combined into a cohesive framework that can predict trade and industrial conditions?

RQ3: How accurately can econometric and machine-learning models predict import dependency, production, and trade flows to assist investment and policy decisions?

RQ4: How can supply-chain risk be measured in a way that is comparable across industries? This includes supplier concentration and critical-input dependency.

RQ5: What are the effects on production, prices, employment, investment, and downstream industries of simulated trade disruptions, foreign export restrictions, and alternative policy interventions?

RQ6: Which decision-support formats are best for converting model output into practical recommendations for manufacturers, legislators, and economic development organizations?

1.6 Significance of the Study

The importance of this study lies in three areas: evidence-based policymaking, industrial competitiveness, and economic security. In terms of the economy, the methodology supports earlier identification of important foreign dependencies and concentration by addressing a proven vulnerability in the U.S. ability to predict trade and manufacturing disruptions rather than just respond to them.

The framework is not intended to be a black box for policy, but rather a tool for decision-making. Its scenario-simulation components, in their fully developed and causally validated form, are meant to enable policymakers to assess the anticipated outcomes of alternative policies, such as tariffs, supplier diversification, and domestic capacity investment, prior to adoption. This longer-term goal outlines a decision-grade causal simulation; the pilot-scale tool shown here (Section 4.6) is an exploratory, non-causal example of that desired capacity.

When combined, these contributions support a national interest in bolstering the analytical framework that underpins U.S. trade and industrial policy. The outputs, which include publications, technical documentation, and an open, replicable methodology, are meant to advance the field beyond the purview of any one project.

1.7 Scope and Delimitations

This study does not suggest any particular tariff, sanction, or regulatory action; rather, it is limited to the development and pilot-scale empirical assessment of a decision-support framework. The framework's structure is industry-neutral, and selected components are evaluated in Chapter Four using 19 actual NAICS-3 manufacturing subsectors that cover a broad range of supply-chain structures, from computer and electronic products to wood products and textile mills. These subsectors were chosen as the set for which comparable annual detail could be constructed across the four federal data sources used in the empirical pilot—BEA, BLS Employment, BLS Productivity, and USITC—between 2015 and 2025.

The research depends on publicly accessible federal datasets and uses recognized economic modeling techniques. The forecasting models are subject to common limitations of econometric

and machine-learning prediction, including sensitivity to data quality, structural change, and unexpected external shocks. The proposed framework addresses these issues through uncertainty analysis, benchmark comparison, transparent reporting, and a planned human-in-the-loop validation process described in Chapter Three and discussed further in Chapter Five.

1.8 Definition of Key Terms

- **Indicators of Industrial Resilience and Competitiveness:** A common set of quantitative metrics that illustrate an industry's susceptibility to supply disruptions, domestic production capabilities, and its relative competitive standing.
- **Supply-Chain Concentration:** The extent to which a crucial input or component is obtained from a limited number of suppliers, countries, or regions, heightens the risk of disruption.
- **Lower-Tier Visibility:** The ability to identify and monitor suppliers beyond the immediate, tier-one relationships, specifically second-, third-, and fourth-tier suppliers.
- **Scenario Simulation:** A modeling approach utilized to assess the economic impact of a potential event, like a tariff, export limitation, or geopolitical disruption, based on defined assumptions.
- **Stress Testing:** The comprehensive assessment of a system's capabilities when faced with unfavorable or extreme situations, adapted here from practices in the financial sector for use in trade and manufacturing systems.
- **Decision-Support Dashboard:** An interactive analytical platform that displays model results, indicators, and scenario outcomes in a format intended for non-technical policy and industry stakeholders.
- **Human-in-the-Loop Validation:** A model-governance approach where subject-matter experts evaluate, interpret, and, if necessary, modify or adjust model results prior to their application in decision-making.

1.9 Organization of the Paper

Chapter One outlines the problems, objectives, research questions, and significance of the study. Chapter Two reviews the relevant literature and positions the framework within that context. Chapter Three presents the five-phase methodology, distinguishing the full proposed architecture from the narrower empirical specification evaluated in this study. Chapter Four presents pilot-scale

empirical results derived from federal data covering 19 manufacturing subsectors from 2015 to 2025. These are empirical estimates based on observed data rather than simulated outcomes, although they represent pilot-scale findings rather than a comprehensive national implementation. Chapter Five discusses the implications, contributions, limitations, recommendations, and directions for future research.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter examines existing literature across four main areas that provide the foundation for the proposed framework. These areas include U.S. trade and industrial data infrastructure, supply-chain visibility and vulnerability, predictive and econometric modeling, and scenario-based policy simulation. The chapter concludes by highlighting the specific gaps in the existing literature that this study aims to address and presents the conceptual framework that guides the research methodology.

2.2 U.S. Trade Policy and Economic Data Infrastructure

The federal government collects and manages a broad range of data relating to trade and industrial activities. Important sources include trade statistics from the U.S. Census Bureau, input-output accounts provided by the BEA, employment and price data from the BLS, trade and tariff information from the U.S. International Trade Commission (USITC), and industrial production indices from the Federal Reserve. Together, these sources represent a significant public investment in the collection and measurement of economic information.

However, an important issue in the use of these data is the gap between the large amount of information available and its practical usefulness for timely analysis across government agencies. Existing classification systems are often presented at a level that is too broad to identify specific dependencies between individual firms or businesses. In addition, the time taken to collect, process, and publish some economic data may be suitable for monitoring broad economic conditions but may not be adequate for identifying and responding to supply-chain risks before they occur. Differences in data collection methods, definitions, and reporting practices among government agencies also make it difficult to combine the information into one consistent analytical system. These challenges support the findings of the U.S. Government Accountability Office (2023), which reported that federal agencies involved in strengthening supply-chain resilience continue to experience difficulties with data collection and the sharing of information across agencies. These limitations therefore provide a strong basis for the data-integration objectives of the framework proposed in this study.

2.3 Supply Chain Visibility and Vulnerability Analysis

Recent disruptions involving semiconductors, critical minerals, and pharmaceutical inputs have led to increased research on supply-chain mapping and the assessment of supply-chain vulnerability. A number of studies have used network-based methods to understand how supply chains are connected. These methods commonly represent supply chains as directed networks, which makes it possible to identify areas where supply is highly concentrated and to examine how disruptions may spread through different parts of the network. Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012) showed that shocks affecting individual firms or sectors can have wider effects on the economy because of the connections created through input-output relationships. This finding supports the focus of the present study on supply concentration as an important pathway through which disruptions can spread across the economy. Similarly, Antràs and Chor (2013) examined ownership and control within global value chains and provided useful evidence for understanding how the structure of international sourcing can indicate exposure to upstream supply risks. Their work supports the idea that the countries from which imports originate can provide useful information about potential supply-chain exposure. However, the country-level Herfindahl-Hirschman Index (HHI) used in this study, as described in Section 4.3, provides a broader measure and cannot capture the detailed firm-level relationships examined in the existing literature.

Visibility beyond the first level of the supply chain remains a major and largely unresolved problem. Although relationships involving tier-one suppliers can generally be identified with reasonable accuracy, the availability and quality of information tend to decline considerably at lower tiers. This is particularly important because disruptions and supply bottlenecks often begin further down the supply chain, where information is more limited. In response to this challenge, the proposed framework combines network-mapping concepts with inference-based approaches, while the empirical pilot uses publicly available data and a country-of-origin concentration proxy rather than a formal firm-level dependency network. In addition, the study draws on the work of Autor, Dorn, and Hanson (2013), whose analysis of competition from Chinese imports is an important contribution to understanding the measurable effects of import competition on domestic manufacturing. Their findings provide a useful basis for examining the relationship between import activity and domestic industrial output, which is further considered in Chapter Four. However, it is important to recognize that Autor, Dorn, and Hanson's (2013) study used an

instrumented causal identification approach that provides stronger evidence of cause and effect than the associational panel regression employed in the present study. The current analysis is based on 19 industry-level cross-sectional units observed over 11 years (209 industry-year observations) and should therefore be interpreted as examining relationships between the variables rather than establishing definitive causal effects.

2.4 Predictive Analytics and Econometric Forecasting in Trade and Manufacturing

Econometric methods have been widely used for many years to forecast trade and manufacturing output. Common approaches include vector autoregression (VAR), error-correction models, and structural gravity models. More recently, machine-learning techniques such as gradient-boosted trees, random forests, and neural network models have gained attention because they can produce more accurate predictions than traditional linear models when sufficient data are available. However, this improvement in predictive performance can come with reduced ease of interpretation. Mullainathan and Spiess (2017) explain that machine learning is mainly concerned with improving prediction rather than identifying and estimating specific model parameters. They also emphasize that achieving high predictive accuracy does not necessarily mean that a model can provide reliable evidence about cause-and-effect relationships, which is often important for policy analysis. The present study follows this distinction by clearly treating the estimated trade coefficients as associational rather than causal, as explained in Section 4.4.

Previous studies have often reported that combining econometric methods with machine-learning techniques can produce better forecasting results than using either approach on its own, especially when the data contains structural changes or major shifts in underlying patterns. This finding provides support for the hybrid approach adopted in this study and discussed in Chapter Three. However, the results presented in Section 4.5 do not show the same outcome. Given the relatively small size of the panel data used in this study, a simple persistence or naive forecasting method performed better than both the linear and machine-learning models. This result shows that findings commonly reported in wider literature may not necessarily apply in the same way to every dataset or research setting.

2.5 Scenario Analysis, Stress Testing, and Policy Simulation

Scenario analysis and stress testing were first widely used in financial risk management but have increasingly been applied to issues relating to trade and industrial policy. Caldara, Iacoviello, Molligo, Prestipino, and Raffo (2020) found that uncertainty surrounding trade policy can reduce investment even before a tariff or other trade measure is actually introduced. This finding is particularly relevant to the present framework because it highlights the importance of assessing the possible effects of proposed policies before they are implemented. Different modeling approaches can provide different perspectives on these potential effects. Computable general equilibrium (CGE) models are useful for examining the broader effects of tariffs and trade agreements across the economy, including effects that may not be captured by partial-equilibrium models. In comparison, agent-based models and network-simulation approaches can complement CGE models by providing greater attention to the behavior and responses of individual firms and the relationships between them.

Existing research generally stresses the importance of evaluating several possible policy interventions within the same analytical framework rather than examining each option separately. Comparing different interventions using consistent methods makes it easier to understand their relative strengths, limitations, and likely effects. This principle is incorporated into the methodology of the present study, particularly in Phase 2 and Phase 3, where alternative policy options are assessed using a common framework.

2.6 Industrial Resilience and Competitiveness Indicators

An increasing body of research has developed composite indicators for measuring industrial resilience. These indicators commonly combine several factors, including the concentration of suppliers, dependence on imports, levels of production capacity utilization, and changes in the workforce and capital base. Existing research generally supports the use of indicators that are clear, consistent, and capable of being measured repeatedly over time, rather than measures that are produced only once. These principles provide the basis for the indicators developed and applied in Phase 1 of the framework used in this study.

This discussion forms part of the recent renewed interest in industrial policy research. Juhász, Lane, and Rodrik (2024) reported that more recent studies using stronger identification methods

tend to find more favorable effects of industrial policy than earlier research, which was generally more doubtful about its benefits. However, their findings also show that the success of industrial policy depends greatly on how policies are designed, implemented, and monitored. In this context, the present framework aims to provide firm- and industry level evidence that can support the development of better-informed industrial policies. The purpose is therefore not to promote industrial policy itself, but to provide useful evidence that can contribute to sound policy design and decision-making.

2.7 Gaps in Existing Literature and Tools

Different methods have already been developed for data integration, supply-chain mapping, forecasting, and scenario-based simulation. However, based on the literature and publicly available federal tools examined in this study, there appears to be limited work on integrated systems that bring all four functions together within a single framework that can be regularly updated and applied across the U.S. manufacturing sector. Many of the existing tools have a narrow focus, depending on proprietary data that are not easily accessible, or mainly examine past events rather than providing forward-looking analysis. This suggests a need for a more integrated and continuously updated framework that can support the identification and assessment of potential manufacturing and supply-chain risks before they occur.

This study responds directly to the identified gap by bringing together four key components: data integration, supply-chain mapping, hybrid forecasting, and comparative scenario simulation. These components are organized within a single modular framework, allowing each part to function as an important element of the overall system. The framework is further assessed through a phased methodological approach, with the aim of developing a system that can be expanded and applied at the national level.

2.8 Conceptual Framework

The conceptual framework developed for this study consists of four connected layers. The first is the data-integration layer, which brings together data obtained from different and often fragmented sources. The second is the mapping layer, which is designed to use network-based methods to represent relationships and dependencies within supply chains. The third is the modeling layer, where econometric methods, machine-learning techniques, and network-risk approaches can be

applied to analyze available data and identify possible risks. The fourth is the decision-support layer, which converts analytical results into practical outputs such as scenarios, indicators, and dashboards. Under the proposed full framework, outputs from each layer would be reviewed through a human-in-the-loop process before informing subsequent stages, in order to assess their quality, relevance, and usefulness.

This structure provides a direct link to the methodology presented in Chapter Three. Phase 1 focuses on the data-integration and mapping layers, while Phase 2 applies the modeling approaches. Phase 3 focuses on developing and testing prototypes to assess the decision-support layer. Finally, Phases 4 and 5 are designed to extend the framework and support its continued updating and application at the national level.

2.9 Summary

This chapter examined existing literature on data infrastructure, supply-chain vulnerability, predictive forecasting, scenario simulation, and indicators of industrial resilience. The review identified a major gap in the literature and available tools examined, particularly the limited development of an integrated national framework that combines these approaches and can be continuously updated. Chapter Three therefore presents the methodology developed in this study to address this identified gap.

CHAPTER THREE: METHODOLOGY

3.1 Research Design

This study adopts a design and development research methodology because the main aim is to develop and empirically assess selected components of an analytical framework rather than to test a single hypothesis. The research process is structured around five sequential but interconnected phases. These are: (i) National Trade and Industrial Intelligence Assessment, (ii) Predictive Economic Framework Development, (iii) Prototype Validation and Pilot Implementation, (iv) National Scaling, Policy Alignment, and Dissemination, and (v) Continuous Economic Intelligence and Improvement. Although the phases follow a planned sequence, they also allow for repeated review and refinement where necessary. This approach supports the gradual development of the framework, ensures that it is empirically tested before wider application, and allows the system to be regularly updated in response to changes in economic and industrial conditions.

3.2 Phase 1: National Trade and Industrial Intelligence Assessment

Phase 1 provides the empirical basis for the study by examining conditions within the U.S. trade and manufacturing sectors, as well as supply chains, investment, labor, energy, and the wider economy. The assessment draws on data from several government sources, including Census trade statistics, BEA input-output and industry accounts, BLS employment and price statistics, USITC trade and tariff data, Federal Reserve production indices, and energy-related data from the Department of Energy (DOE). These data are used to assess the performance of selected industries and identify major areas of vulnerability, production capacity limitations, supply-chain risks, and challenges affecting their competitiveness.

Alongside this assessment, the study also reviews and organizes existing federal datasets and tools related to industrial resilience. These resources are assessed based on factors such as the level of detail they provide, how frequently they are updated, their ability to work with data from other systems, and the extent to which they provide information about lower levels of the supply chain. This assessment helps to identify the specific limitations in the existing data infrastructure that the integration activities in Phase 2 will need to address.

A major output of Phase 1 is the development of a supply-chain dependency map. The map shows the relationships between foreign suppliers, imported inputs, domestic manufacturing firms, and downstream industries. These relationships are represented as a directed network, with the connections weighted according to the value of transactions and, where sufficient information is available, the level of supplier concentration. This approach provides a clearer picture of how different parts of the supply chain depend on one another.

The phase ends with the development of baseline Industrial Resilience and Competitiveness Indicators. These indicators cover important areas such as supplier concentration, dependence on imports, capacity utilization, availability of workers, and trends in capital investment. The findings from these indicators are then used to identify and rank the industries that require further assessment through the predictive modeling activities planned for Phase 2.

3.3 Phase 2: Predictive Economic Framework Development

Phase 2 focuses on developing the integrated framework by bringing together the data generated and organized during Phase 1 and applying four main categories of models.

The forecasting models use a combination of econometric and machine-learning methods, as discussed in Section 2.4. VAR and error-correction models are used to examine the structural relationships among the variables, while gradient-boosted tree models and panel-based machine-learning methods are applied to identify nonlinear relationships and interactions that may not be adequately captured by conventional econometric models. The performance of the different forecasting approaches is then compared using out-of-sample results. Where appropriate, the forecasts may be combined using ensemble weighting, with greater weight given to models that demonstrate stronger predictive performance outside the estimation sample.

The supply-chain risk models are designed to measure the extent to which industries depend on a limited number of foreign suppliers. Supplier concentration is assessed using measures such as the HHI, which is also used in U.S. antitrust and merger-review practice (U.S. Department of Justice & Federal Trade Commission, 2023). In this study, the HHI is applied to patterns of import sourcing to assess the degree of concentration among supplier countries. These measures are combined with network centrality indicators derived from the supply-chain dependency map

developed in Phase 1. This makes it possible to assess not only how concentrated import sources are, but also how important particular suppliers are within the wider supply-chain network.

The economic-impact models are used to assess how a disruption may spread through the supply-chain dependency network and affect important economic outcomes. These outcomes include production, prices, employment, investment, and activities in downstream industries. The analysis builds on input-output multiplier methods and incorporates elasticity estimates obtained during the forecasting stage to provide a better understanding of how changes in one part of the supply chain may influence other areas of the economy.

The scenario, stress-testing, and policy-simulation models bring together the different modeling approaches developed in the earlier stages. They are used to examine the possible effects of specific events and policy options under clearly defined conditions. The scenarios considered include tariffs, restrictions on exports, geopolitical disruptions, supply shortages, and changes in demand. The framework also considers alternative policy responses, such as investment in domestic production capacity and diversification of suppliers. The different scenarios are compared using a common set of assumptions so that their potential economic effects can be assessed consistently.

Under the proposed Phase 2, several quality-control measures would be applied to improve the reliability of the results. These would include checking data quality, assessing uncertainty, documenting modeling procedures, and comparing model performance with simple forecasting methods and existing baseline models. Outputs intended for decision support would also be reviewed by subject-matter experts through the planned human-in-the-loop validation process described in Section 3.9.1. This review would occur before such outputs are used in decision-support applications, helping to assess whether the findings are reasonable, understandable, and relevant for policy and economic decision-making.

3.4 Phase 3: Prototype Validation and Pilot Implementation

The proposed Phase 3 focuses on pilot testing and validation of the framework using priority industries identified during Phase 1. At this stage, data assembled in the earlier phases would be used to develop and evaluate predictive models, supply-chain vulnerability measures, scenario-

testing applications, and dashboards for presenting results in a clear and useful form. The empirical work in Chapter Four implements selected elements of this phase, but it does not constitute full Phase 3 validation.

Under the proposed full Phase 3, validation would be carried out in two main ways. First, retrospective validation would apply the framework to previous disruption events, such as earlier tariff measures, export-control events, and supply shortages, and compare model outputs with observed outcomes. Second, prospective stress testing would focus on potential risks that have not yet occurred, generating forward-looking assessments that could be reviewed and updated as economic and supply-chain conditions change. The current pilot includes pseudo-out-of-sample forecast evaluation and robustness testing, but broader event-based retrospective validation remains future work.

In the empirical pilot, forecasting performance is evaluated using standard measures, including Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), together with comparison against a naive persistence benchmark. Under the proposed full Phase 3, sensitivity analysis and structured subject-matter-expert feedback would provide additional validation. Findings from those broader validation activities would be used to refine the models before application to additional industries.

At the end of the proposed full Phase 3, the framework would produce standardized data-integration procedures, model-specification templates, and validation checklists. These resources are intended to support consistent and repeatable application when the framework is extended to other industries and geographic regions.

3.5 Phase 4: National Scaling, Policy Alignment, and Dissemination

The proposed Phase 4 focuses on expanding framework components that have been sufficiently pilot-tested beyond the industries included in the initial study to other strategically important sectors and regional economies. The modular design is intended to allow new industries or regions to be incorporated without redesigning the entire framework. At this stage, standardized national indicators for industrial resilience, international-trade exposure, production capacity, supplier

concentration, and competitiveness would be further developed and evaluated for meaningful comparison across industries and regular updating.

The scaling process involves collaboration with relevant federal government agencies, universities, professional organizations, and regional economic-development stakeholders. Their involvement helps to test the framework under a wider range of economic and industrial conditions while also ensuring that the information produced meets the practical needs of those who may use it. The findings from this phase are shared through academic publications, technical reports, policy briefs, conferences, and workshops. In addition, implementation guides are prepared to explain how the framework can be replicated and applied consistently in other industries and regions.

3.6 Phase 5: Continuous Economic Intelligence and Improvement

The proposed Phase 5 is designed to ensure that the framework remains dynamic and regularly updated rather than functioning as a fixed or one-time system. New data would be incorporated at regular intervals, forecasts would be compared with subsequently observed outcomes, and models would be recalibrated when economic and industrial relationships change. In this way, retrospective validation would become part of ongoing operation, helping to identify and reduce model drift over time.

In addition, Phase 5 introduces regular stress-testing exercises at both the national and sectoral levels. The methods used within the framework are reviewed and updated as new sources of data, analytical techniques, and modeling approaches become available. This continuous process of assessment and improvement helps the framework remain useful and relevant as economic, industrial, and supply-chain conditions change.

3.7 Data Sources

The framework relies mainly on data from several U.S. federal sources. These include Census trade and manufacturing statistics, BEA input-output and gross-output-by-industry data, BLS information on employment, wages, and prices, USITC data on tariffs and trade-remedy measures, Federal Reserve production indices, and energy-related data from the DOE. Where relevant information is available, data from industry associations are also included to provide additional insight into lower levels of the supply chain that may not be adequately captured by federal

statistics alone. To support transparency and make the research process reproducible, all data sources used in the study are recorded together with their extraction dates and details of any transformations or processing carried out on the data. The procedures for documenting and managing this data are further described in Section 3.3.

3.8 Modeling Approach

The modeling strategy used in this study brings together three main approaches. The first is econometric time-series modeling, which includes VAR, based on Sims (1980), error-correction models, and structural gravity models associated with Anderson (1979). These methods are useful because they provide interpretable results that can support economic and policy-related forecasting. The second approach involves machine-learning methods, particularly gradient-boosted trees, following Friedman (2001), and random forests, introduced by Breiman (2001). These methods are included because they can identify nonlinear relationships and patterns that may not be fully captured by traditional econometric models. The forecasts from the machine-learning models are compared with those from the econometric models and, where appropriate, combined using ensemble weights determined by their performance on data outside the model estimation period. The third approach uses network and input-output methods, drawing on the work of Leontief (1936) and the applied general-equilibrium tradition of Shoven and Whalley (1984) for analyzing relationships across sectors. These methods allow the structure of the supply chain to be represented more explicitly and help trace how a disruption affecting one part of the network may spread to other connected firms, industries, or sectors. By combining these three approaches, the study is able to consider both the measurable economic relationships between variables and the wider network effects that may arise when disruptions occur within interconnected supply chains.

3.8.1 Panel Data Model Specification

The main econometric component of the study is based on a panel data model, rather than relying solely on cross-sectional or time-series data. The use of panel data is appropriate for several reasons. First, it allows the study to account for differences between industries that do not change significantly over time and may not be directly observed in the data. Second, it makes it possible to control for economic shocks that affect several or all industries during the same period. Third,

combining observations across industries and over time provides a larger number of usable observations than would normally be available from a purely cross-sectional or time-series research design.

The study presents two separate model specifications. These are deliberately distinguished from each other to ensure that the broader architecture proposed for the framework is not confused with the more limited empirical model that is actually estimated in the pilot analysis presented in Chapter Four.

Proposed National Framework Specification

The framework's full proposed specification, intended for eventual national-scale implementation once richer input-cost, tariff, and exchange-rate data are integrated, is a two-way (entity- and time-) fixed effects model:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 \text{Tariff}_{it} + \beta_2 \text{HHI}_{it} + \beta_3 \ln(\text{ForeignInputPrice})_{it} + \beta_4 \text{CapUtil}_{it} + \beta_5 \ln(\text{LaborCost})_{it} + \beta_6 \ln(\text{EnergyPrice})_{it} + \beta_7 (\text{ImportShare}_i \times \text{RER}_t) + \varepsilon_{it}$$

where i indexes industry and t indexes year; α_i and λ_t are entity and time fixed effects; Y_{it} is log domestic production; Tariff_{it} is the applicable tariff rate; HHI_{it} is import-origin concentration; $\text{ForeignInputPrice}_{it}$, LaborCost_{it} , and EnergyPrice_{it} are input-cost indices; and $\text{ImportShare}_i \times \text{RER}_t$ captures currency-movement exposure. This requires tariff, foreign-input-price, capacity, and exchange-rate data beyond this study's four integrated sources and is not the specification estimated in Chapter Four.

Empirical Pilot Specification Estimated in This Study

Chapter Four estimates a narrower specification, using only variables constructible from the four integrated federal sources:

$$\log(Y_{it}) = \alpha_i + \lambda_t + \beta_1 \text{Employment}_{it} + \beta_2 \text{LaborProductivity}_{it} + \beta_3 \log(\text{Imports}_{it}) + \beta_4 \log(\text{Exports}_{it}) + \beta_5 \text{HHI}_{it} + \varepsilon_{it}$$

where Y_{it} is Gross Output (BEA), Employment_{it} and $\text{LaborProductivity}_{it}$ are from BLS, Imports_{it} and Exports_{it} are from USITC, and HHI_{it} is the Import-Origin Concentration Index (Section 3.3). This is the exact specification estimated in Table 4.2 and used for the main panel-regression analysis in Section 4.4; other Chapter Four analyses, such as forecasting and scenario illustrations, use the methods described in their respective sections.

Model selection between fixed- and random-effects specifications is governed by a like-for-like Hausman (1978) test, supplemented with a Mundlak (1978) correlated random effects test (Section 4.4.1). Standard errors are entity-clustered throughout, with a wild-cluster bootstrap (Cameron, Gelbach, & Miller, 2008) given this pilot's modest 19 clusters (Section 4.4.2), following Petersen (2009).

Because manufacturing output likely exhibits genuine persistence through capacity, inventory, and order-backlog dynamics, a dynamic extension adds a lagged dependent variable:

$$\log(Y_{it}) = \rho \log(Y_{i,t-1}) + \beta'X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

This introduces a well-documented small-sample bias (Nickell bias) in short panels. Arellano-Bond (1991) or Blundell-Bond (1998) GMM are the standard solutions in principle but are not applied here: with only 19 cross-sectional units, GMM risks instrument proliferation and unreliable overidentification tests. Section 4.4.5 instead reports a simpler dynamic fixed-effects specification, explicitly flagged as subject to Nickell bias, as a parsimonious check rather than a bias-corrected estimate; full GMM is future work contingent on a larger panel (Section 5.6).

3.8.2 Predictive Modeling Approach

The forecasting component of the framework uses a pseudo-out-of-sample, rolling-origin approach to generate one-year-ahead predictions. Under this procedure, all predictor variables are lagged by one year, and the models are repeatedly re-estimated at different points in time using only the information that would have been available up to each forecasting point. The resulting predictions are then tested against the actual outcome observed in the following year. The approach is described as pseudo-out-of-sample rather than fully ex-ante because it uses the current version of historical data, which may include later revisions, rather than the exact data version that would have been available when each historical forecast was originally made. This distinction is stated clearly to improve methodological transparency.

The forecasting origins are moved forward across the panel, with each model trained using data up to a particular year and then evaluated using the outcome for the following year. This produces both an overall average performance measure and results for individual years. Reporting the year-specific results is important because it allows the reader to determine whether the overall performance is being influenced heavily by one particular year. Four forecasting approaches are

compared: naive persistence, a linear model with entity fixed effects and lagged predictors, a gradient-boosted ensemble as described in Section 2.4, and a hybrid model that combines the forecasting results.

Model performance is assessed using three standard forecasting measures: MAPE, RMSE, and Mean Absolute Error (MAE). The results are also compared with the naive persistence benchmark, as described in Section 3.9. Including this simple benchmark is an important part of the methodology because it provides a basic standard against which the more complex models can be judged. If a more advanced forecasting approach cannot perform better than simple persistence, it would be difficult to claim that the approach provides meaningful additional forecasting value. Therefore, the study reports the benchmark results even in cases where the naive method performs as well as, or better than, the more complex models.

3.9 Validation and Evaluation Metrics

Forecasting performance is assessed by comparing the predicted values with the actual outcomes using standard measures such as MAPE and RMSE. The performance of the models is also compared with simple naive forecasts and, where such information is available, with forecasts that have already been published by other recognized sources. This provides a clearer basis for determining whether the models used in the study offer useful improvements in forecasting accuracy.

The supply-chain risk indicators are assessed by comparing the inputs identified by the models as high-risk with supply-chain disruptions that have been documented in the past. This helps to determine whether the indicators are capable of identifying areas that have previously experienced significant disruption. Similarly, the outputs generated from the scenario analysis are tested retrospectively against previous disruption events, as described in Section 3.4, to examine how well the framework reflects outcomes observed during actual events.

All findings are recorded transparently, including cases where the models perform poorly or fail to improve on the selected benchmarks. Reporting these limitations forms an important part of the model-governance process and helps ensure that the strengths and weaknesses of the framework are clearly documented.

3.9.1 Human-in-the-Loop Validation Protocol

This section describes the human-in-the-loop validation protocol introduced earlier as one of the key design principles of the framework. The protocol is proposed for implementation during the pilot stage and would depend on the institutional collaboration outlined in Section 3.5. Therefore, the procedure described in this section represents a planned specification for future implementation rather than a review process that is currently in operation.

Under the proposed system, every output intended for external use would be reviewed by two different roles. The first would be a methods reviewer, whose responsibility would be to check whether the model specifications, diagnostic procedures, and decisions made during data construction are technically appropriate. The second would be a domain reviewer, such as an economist, policy analyst, or manufacturing subject-matter expert. This reviewer would assess whether the results make practical and economic sense, including aspects that may not be adequately captured by the statistical models alone.

Three main situations would require an output to undergo review before it could be released. The first would occur when a forecast exceeds a predetermined magnitude threshold in relation to the level of historical volatility. The second would apply when an updated regression produces substantial changes in its coefficients compared with the previous version, consistent with the specification comparisons presented in Sections 4.4.1–4.4.6. The third would occur when the result relates to a subsector experiencing a known disruption at the same time, since the assumptions of stationarity are more likely to be affected under such conditions.

Each reviewed output would be recorded together with the reviewer's identity, the date of the review, and the reason for the decision or assessment. This would create a clear and auditable record and support the framework's commitment to open-source reproducibility, as discussed in Section 5.8. The proposed process is intentionally less complex than the formal model-risk management systems commonly used in the financial sector. It is designed to be practical for a research pilot team with limited resources. However, if the framework is expanded to national-level implementation in Phase 4, a more formal and comprehensive governance structure would be required.

3.10 Ethical Considerations and Limitations

The framework uses only data that are publicly available or obtained through appropriate licensing arrangements. Where industry-level information is used for validation purposes, the data are presented in an aggregated or anonymized form to protect the identity of individual organizations. The framework is intended to function as a decision-support tool rather than an automated system that independently makes policy decisions. Its purpose is therefore to provide evidence that can assist decision-makers, not to replace professional or expert judgement. In line with this approach, the human-in-the-loop process proposed for Phases 2 and 3 requires relevant outputs to be reviewed by experts before they are used to support decision-making.

From a methodological perspective, the framework also has some of the limitations commonly associated with econometric and machine-learning forecasting. Its performance may be affected by the quality and completeness of the available data, while forecasting accuracy may decline when unexpected structural changes occur that are not captured by the models. In addition, all forward-looking forecasts contain a degree of uncertainty because future economic and industrial conditions cannot be known with certainty. Although these limitations cannot be completely removed, the framework seeks to reduce their effects through uncertainty analysis, regular model review, and continuous recalibration during Phases 2, 3, and 5. These limitations and their implications for the study are considered in greater detail in Chapter Five.

CHAPTER FOUR: EMPIRICAL RESULTS

4.1 Introduction

This chapter reports empirical results generated entirely from real, primary-source federal data, following the methodology described in Chapter Three. A panel dataset covering 19 NAICS-3 manufacturing subsectors observed annually from 2015 through 2025 ($N = 209$ industry-year observations) was assembled from four federal sources: BEA (Gross Output by Industry), BLS (Current Employment Statistics and Industry Productivity), and USITC (DataWeb imports, exports, and country-of-origin data used to construct an Import-Origin Concentration Index). Full source citations and construction notes are in the accompanying data workbook and Appendix A.

The forecasting exercise (Section 4.5) is described as pseudo-out-of-sample rather than genuinely ex-ante, since it uses today's data vintage rather than the vintage available on each historical forecast date; the Hausman test (Section 4.4.1) is estimated like-for-like; inference for the main regression includes a wild-cluster bootstrap given the panel's modest number of clusters (Section 4.4.2); and the concentration variable is named precisely to describe what it measures. Every result, significant or not, is reported directly.

Section 4.2 documents data-infrastructure gaps (RQ1). Section 4.3 reports descriptive statistics (RQ2). Section 4.4 reports the main panel regression and its robustness and diagnostic extensions (RQ2, RQ4). Section 4.5 reports the forecasting exercise (RQ3). Section 4.6 reports the scenario analysis (RQ5). Section 4.7 describes the decision-support interface (RQ6). Section 4.8 maps these results against the five-phase methodology from Chapter Three.

4.2 Documented Data-Infrastructure Gaps

Consistent with RQ1, assembling the panel directly surfaced the data-infrastructure limitations identified in Section 2.2, reported here as an empirical finding rather than a methodological aside.

Four gaps were documented. First, no single federal source integrates output, employment, productivity, and trade; each variable required a separate query against a separate agency system (BEA, BLS's two separate programs, USITC), with no shared access point. Second, classification granularity is inconsistent: BEA and BLS report some NAICS-3 categories only in combined form (e.g., Food and Beverage/Tobacco, NAICS 311/312), while USITC trade data is finer-grained, requiring upward aggregation and a documented approximation (Appendix A). Third, USITC reports Transportation Equipment (NAICS 336) as one combined trade figure, while BEA/BLS

separate Motor Vehicles from Other Transportation Equipment, requiring a second documented approximation (employment-share splitting). Fourth, the country-of-origin detail needed for concentration analysis required a substantially larger, separate query (nearly 40,000 rows) than headline trade statistics, indicating that lower-tier visibility, while technically available, requires deliberate additional effort.

These gaps answer RQ1 directly: the limitation is fragmentation and inconsistent granularity, not an absence of underlying data. Each was resolved here through disclosed construction decisions, but each represents a standing barrier any national-scale implementation would need to resolve systematically rather than ad hoc.

4.3 Panel Dataset and Descriptive Statistics

Table 4.1 reports descriptive statistics across all 209 observations, addressing RQ2. The panel spans subsector sizes from \$15.7B (Apparel) to \$1,268.3B (Food and Beverage/Tobacco) in gross output, and import-origin concentration from HHI 646 to 3,606. A full year-by-subsector snapshot is in Appendix A.

Table 4.1. Descriptive Statistics, Full Panel ($N = 209$; 19 NAICS-3 Manufacturing Subsectors $\times$ 11 Years, 2015–2025)

Statistic	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	Import-Origin HHI (0–10,000)
Mean	330.1	658.1	76.0	124.6	97.9	1,558.4
Std. Deviation	287.5	462.3	72.3	115.0	6.0	672.9
Minimum	15.7	79.0	4.4	4.9	78.6	645.9
Median	231.4	427.4	59.4	94.9	99.3	1,481.8
Maximum	1,268.3	2,098.7	302.7	709.4	115.0	3,605.9

This variable is termed the Import-Origin Concentration Index rather than "Supplier Concentration" to describe precisely what it measures: concentration of import value among countries of origin (Herfindahl-Hirschman methodology, Section 3.3), computed from USITC data. It is a macro-level, country-of-origin proxy, not a firm-level supplier measure, and does not specifically isolate critical inputs, it reflects all trade value classified within a subsector's USITC codes. In 2025, Food and Beverage/Tobacco showed the lowest concentration (HHI 837); Wood Products and Printing showed the highest (above 1,700).

A measurement caveat: Gross Output, Imports, and Exports are in nominal (current) dollars, not deflated. Year fixed effects absorb economy-wide inflation but not subsector-specific price

movements. This is disclosed as a limitation rather than corrected here; a real (deflated) panel is a priority for future work (Section 5.6).

4.4 Panel Regression: Explaining Manufacturing Output

The two-way fixed-effects specification (Section 3.8.1) was estimated with $\log(\text{Gross Output})$ as the dependent variable and Employment, Labor Productivity, $\log(\text{Imports})$, $\log(\text{Exports})$, and standardized Import-Origin Concentration as regressors. Standard errors are entity-clustered; because the panel has only 19 clusters, a wild-cluster bootstrap (Section 4.4.2) is the more credible inferential statement than the asymptotic p-values below.

Table 4.2. Two-Way Fixed Effects Panel Regression: Dependent Variable: $\log(\text{Gross Output})$; $N = 209$; entity-clustered standard errors (asymptotic; see Section 4.4.2 for wild-cluster-bootstrap inference)

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value
Employment (thousands)	0.0002	0.0002	0.85	0.397 (n.s.)
Labor Productivity (Index, 2017=100)	0.0005	0.0018	0.30	0.765 (n.s.)
$\log(\text{Imports})$	0.1998	0.1023	1.95	0.052
$\log(\text{Exports})$	0.2747	0.1349	2.04	0.043
Import-Origin Concentration (HHI, standardized)	-0.0240	0.0220	-1.09	0.275 (n.s.)

Model fit: R^2 (within) = 0.619; $F(5, 175) = 17.96$, $p < 0.001$. Under asymptotic clustering, exports are significant and imports marginally so; under the wild-cluster bootstrap (Section 4.4.2), neither reaches significance (imports $p = 0.142$; exports $p = 0.078$). The honest conclusion is more conservative: trade activity is positively associated with output, but this panel does not provide statistically decisive evidence of that association under small-sample-appropriate inference. These coefficients reflect co-movement, not a causal trade elasticity. Employment and Labor Productivity are not significant; Import-Origin Concentration is negative but not significant under either inference approach ($p = 0.275$ asymptotic, $p = 0.263$ bootstrap).

4.4.1 Robustness Checks: Alternative Model Specifications and a Like-for-Like Hausman Test

Four alternative specifications were estimated on the identical panel: Pooled OLS, Random Effects, Two-Way FE excluding concentration, and Entity-FE-only.

Table 4.3. Alternative Model Specifications, Dependent Variable: $\log(\text{Gross Output})$; $N = 209$

Variable	Pooled OLS	Random Effects	Two-Way FE (no HHI)	Two-Way FE (Main Model)	Entity FE only
Employment	0.0007**	0.0005**	0.0001 (n.s.)	0.0002 (n.s.)	0.0003 (n.s.)
Labor Productivity	−0.0193**	−0.0013 (n.s.)	0.0007 (n.s.)	0.0005 (n.s.)	−0.0010 (n.s.)
log(Imports)	−0.5432**	0.1983*	0.1934*	0.1998*	0.2824**
log(Exports)	0.8430**	0.4078**	0.2526*	0.2747**	0.3611**
Import-Origin Concentration (HHI, std.)	−0.0721*	−0.0764**	n/a	−0.0240 (n.s.)	−0.0679**
R ² (overall / within)	0.861	0.693 / 0.704	0.329 / 0.553	0.339 / 0.619	0.710

A Hausman test comparing Random Effects against Fixed Effects is only valid on a like-for-like basis, both models must share the same year-effect structure. Table 4.4 reports a corrected, like-for-like Hausman test (Random Effects re-estimated with year dummies) alongside a Mundlak (correlated random effects) test.

Table 4.4. Model-Selection Tests: Fixed Effects vs. Random Effects

Test	Statistic	p-value	Conclusion
Hausman test (like-for-like, year effects in both models)	$\chi^2(5) = 5.36$	0.374	Fails to reject Random Effects
Mundlak test (joint significance of entity-mean regressors)	$\chi^2(5) = 48.97$	< 0.0001	Rejects Random Effects; supports Fixed Effects

These tests point in different directions, and both are reported. The Hausman test alone does not decisively favor Fixed Effects. The Mundlak test, which directly tests whether entity-level averages are correlated with the unobserved individual effect, does support Fixed Effects, and is treated as the more directly relevant test here. Fixed Effects remains the preferred specification as a reasoned judgment from mixed evidence, not a test result that settles the question beyond doubt.

The Import-Origin Concentration coefficient is negative and significant in three of the four alternative specifications, losing significance only in the preferred two-way fixed-effects model. It would be tempting to treat this as evidence the relationship is robust but once two-way FE is identified as the preferred specification, significance in less-controlled alternatives does not strengthen the main result; if anything, losing significance once common year shocks are controlled for suggests the apparent relationship in weaker specifications may partly reflect shared trends. The honest conclusion: Import-Origin Concentration has a consistently negative coefficient

across specifications, but the preferred specification does not provide statistically conclusive evidence of a within-industry effect.

4.4.2 Additional Diagnostics

Additional diagnostic tests were run on the real panel and are reported regardless of outcome.

Table 4.5. Additional Diagnostic Tests

Diagnostic	Result	Interpretation
Wild-cluster bootstrap (999 reps, cluster = entity), log(Imports)	statistic = 1.91, $p = 0.142$	Not significant at 5%; weaker than the asymptotic result
Wild-cluster bootstrap, log(Exports)	statistic = 1.99, $p = 0.078$	Not significant at 5%, marginal at 10%
Wild-cluster bootstrap, Import-Origin Concentration	statistic = -1.07 , $p = 0.263$	Not significant, consistent with asymptotic
Variance Inflation Factors (all regressors)	range 1.08–4.41	Well below conventional concern thresholds
Pesaran CD test (cross-sectional dependence)	CD = -1.38 , $p = 0.169$	Not significant; supports entity-only clustering
Wooldridge-style serial correlation test (H_0 : coefficient = -0.5)	$t = 8.65$, $p < 0.0001$	Rejects the null; evidence of persistence a static model misses

Three findings matter. First, the wild-cluster bootstrap is more credible than large-cluster asymptotic for a 19-entity panel, finds neither trade coefficient significant at 5%, revising the paper's conclusions accordingly: trade activity is positively associated with output, but not decisively so under appropriate inference. Second, the absence of cross-sectional dependence supports the entity-only clustering used throughout. Third, the serial correlation test rejects its null, indicating persistence a static model misses, addressed directly with a dynamic robustness check in Section 4.4.5.

4.4.3 Transportation-Equipment Allocation Sensitivity Check

Section 4.2 discloses an approximation: USITC's combined NAICS 336 trade figure is split between Motor Vehicles and Other Transportation Equipment by employment share. Table 4.6 re-estimates the main specification three ways; current allocation, omitting both affected subsectors, and collapsing them into one category matching USITC's native reporting.

Table 4.6. Main Specification Under Three Transportation-Equipment (NAICS 336) Allocation Approaches

Variable	Current allocation (main, N=209)	Omit both transport subsectors (N=187)	Collapsed to one category (N=198)
log(Imports)	0.1998 (p=0.052)	0.2064 (p=0.071)	0.2008 (p=0.059)
log(Exports)	0.2747 (p=0.043)	0.2809 (p=0.072)	0.2720 (p=0.058)
Import-Origin Concentration (HHI)	-0.0240 (p=0.275)	-0.0286 (p=0.292)	-0.0257 (p=0.282)

All three coefficients remain close in sign and magnitude; the transportation-equipment allocation approximation does not appear to drive the paper's results.

4.4.4 Trade × Concentration Interaction

The baseline model cannot distinguish whether concentrated subsectors are more vulnerable to a given trade shock than diversified ones. Table 4.7 tests this directly with an Imports × Concentration interaction term.

Table 4.7. Two-Way Fixed Effects Model with Trade × Concentration Interaction

Variable	Coefficient	Std. Error	t-stat	p-value
Employment	0.0003	0.0002	1.08	0.280 (n.s.)
Labor Productivity	0.0011	0.0016	0.64	0.524 (n.s.)
log(Imports)	0.2018	0.1033	1.95	0.052
log(Exports)	0.2933	0.1386	2.12	0.036
Import-Origin Concentration (HHI, std.)	-0.3207	0.6642	-0.48	0.630 (n.s.)
log(Imports) × Concentration (HHI, std.)	0.0119	0.0265	0.45	0.653 (n.s.)

The interaction term is not significant ($p = 0.653$): this panel does not find evidence that highly concentrated subsectors are more sensitive to import fluctuations than diversified ones. The absence of significance may reflect limited statistical power, model specification, measurement limitations, or a genuine absence of the effect this panel cannot distinguish among these explanations, and re-testing with a larger panel is a priority (Section 5.6).

4.4.5 Dynamic Panel Robustness Check

Section 4.4.2's serial-correlation diagnostic found persistence a static model miss. Table 4.8 reports a parsimonious dynamic check: the specification extended with a one-period lagged dependent variable. This is explicitly not an Arellano-Bond or Blundell-Bond GMM estimate (with only 19 entities, GMM risks instrument proliferation and is subject to a well-documented small-sample bias, Nickell bias, in short panels), reported as a robustness check, not a bias-corrected structural estimate.

Table 4.8. Dynamic Panel Specification with Lagged Dependent Variable ($N = 190$; one year lost to lagging); asymptotic entity-clustered inference

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value	Wild-bootstrap p-value
log(Gross Output), lagged one year	0.4149	0.1927	2.15	0.033	0.063
Employment	0.0001	0.0002	0.83	0.408 (n.s.)	0.405 (n.s.)
Labor Productivity	0.0009	0.0016	0.58	0.561 (n.s.)	0.613 (n.s.)
log(Imports)	0.2113	0.1176	1.80	0.074	0.233 (n.s.)
log(Exports)	0.1270	0.1080	1.18	0.241 (n.s.)	0.256 (n.s.)
Import-Origin Concentration (HHI, std.)	-0.0257	0.0196	-1.32	0.190 (n.s.)	0.260 (n.s.)

The lagged output term's asymptotic coefficient ($\rho = 0.41$) is consistent with substantial persistence, but under the small-sample-appropriate wild-cluster bootstrap it is only marginally significant ($p = 0.063$ vs. 0.033 asymptotically) suggestive but not decisive evidence of persistence.

Imports and concentration move only modestly from the static specification (Table 4.2), but the export coefficient falls from 0.2747 to 0.1270, roughly half its static value, losing its marginal significance. This suggests part of the static model's export-output association reflects output's own persistence rather than a stable contemporaneous relationship. Both specifications are reported so the reader can see how sensitive the finding is to this choice.

4.4.6 First-Differenced Robustness Check

Output, Imports, and Exports are nominal, not deflated (Section 4.3); since all three trend upward, part of the main specification's associations could, in principle, reflect shared trends. As an additional check, the specification was re-estimated in first differences. To be precise, differencing

alone removes each entity's fixed effect, but an entity-specific linear trend survives as an entity-specific constant. Removing it requires including entity dummies in the differenced regression itself, which this specification does. A wild-cluster bootstrap is again reported alongside the asymptotic result.

Table 4.9. First-Differenced Specification with Entity and Year Effects ($N = 190$; one year lost to differencing)

Variable	Coefficient (main FE, Table 4.2)	Coefficient (first-differenced)	Asymptotic p-value	Wild-bootstrap p-value
Employment	0.0002	-0.0001	0.896 (n.s.)	0.931 (n.s.)
Labor Productivity	0.0005	0.0015	0.193 (n.s.)	0.161 (n.s.)
log(Imports)	0.1998	0.2124	0.294 (n.s.)	0.556 (n.s.)
log(Exports)	0.2747	0.2499	0.019	0.001
Import-Origin Concentration (HHI, std.)	-0.0240	-0.0447	0.225 (n.s.)	0.503 (n.s.)

With entity-specific trends properly removed, imports lose significance entirely under both inference approaches. Exports remain significant, close in magnitude to the main result (0.2747 vs. 0.2499), and notably under the wild-cluster bootstrap ($p = 0.001$), the only trade coefficient in this chapter that remains significant under small-sample-appropriate inference once trends are controlled for. This does not resolve the nominal-versus-real measurement limitation (Section 4.3), but identifies the export-output association specifically, not the trade-output association generally, as the most robust single finding in this chapter.

4.4.7 Employment-Impact Model

Sections 4.4–4.4.6 focus on output. This section extends the framework specifically to employment while controlling for output and productivity, directly addressing the goal of estimating disruption effects on production and employment rather than output alone.

Table 4.10. Employment-Impact Model: Dependent Variable: $\log(\text{Employment})$; $N = 209$; entity-clustered standard errors

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value	Wild-bootstrap p-value
log(Gross Output)	0.1322	0.1051	1.26	0.210 (n.s.)	0.281 (n.s.)
Labor Productivity	-0.0041	0.0025	-1.61	0.109 (n.s.)	0.213 (n.s.)
log(Imports)	0.1467	0.0832	1.76	0.080 (n.s.)	0.131 (n.s.)
log(Exports)	-0.0287	0.0983	-0.29	0.770 (n.s.)	0.820 (n.s.)
Import-Origin Concentration (HHI, std.)	0.0346	0.0217	1.60	0.112 (n.s.)	0.204 (n.s.)

None of the five regressors reach significance under either inference approach, a genuine null result, reported directly. One pattern worth noting despite the lack of significance: the concentration coefficient is positive here, the opposite sign from its output association, consistent with a plausible (but statistically unconfirmed) story that more concentrated subsectors may be somewhat more labor-intensive per unit of output. Establishing genuine employment-impact estimates is a priority for a larger panel (Section 5.6).

4.5 Predictive Model: Pseudo-Out-of-Sample Forecast Evaluation

A forecasting exercise that uses a period's actual realized predictor values to predict that period's output is conditional prediction (nowcasting), not ex-ante forecasting, since those values would not genuinely have been available at forecast time. This exercise avoids that by lagging every predictor one year, so only information available at $t-1$ predicts t .

The model is retrained at each of four origin years (2021–2024) using only data available up to that point, forecasting exactly one year ahead each time, a rolling-origin design (see Section 3.8.2 for why "pseudo-out-of-sample" rather than "genuinely ex-ante" is the precise term, given current rather than historical data vintages are used). Concentration is standardized using only each origin's training window. Four approaches are compared: naive persistence; linear (entity FE + trend, lagged); gradient boosting; and a hybrid average.

Table 4.11. Pseudo-Out-of-Sample Rolling-Origin Forecast Accuracy (MAPE), by Individual Forecast Year and Average

Model	2022	2023	2024	2025	Average MAPE	Avg. RMSE (\$B)	Avg. MAE (\$B)
Naive persistence	10.19%	5.06%	2.53%	1.97%	4.94%	37.43	20.68
Linear (entity FE + trend, lagged)	13.44%	6.34%	6.18%	6.22%	8.04%	49.85	30.18
Gradient Boosting (lagged)	15.70%	14.13%	12.80%	4.99%	11.90%	118.91	58.05
Hybrid (Linear + GBR average)	12.18%	9.86%	7.61%	4.40%	8.51%	82.66	40.80

The naive persistence benchmark achieves the lowest average error of all four approaches by a wide margin (4.94% vs. 8.04% or worse for every model-based approach), reported directly. With

genuinely available lagged predictors at this panel's size, neither the econometric nor the machine-learning approach adds forecasting value over simply carrying forward each subsector's most recent output.

This carries a direct implication for the proposed framework: one-year-ahead forecasting with this panel's covariates is genuinely difficult, and a deployed version should be benchmarked against, and should expect to sometimes underperform, a simple persistence forecast precisely the evaluation discipline specified in Section 3.9.

4.6 Model-Implied Scenario Analysis: Illustrative, Non-Causal Trade Disruption Effects

This section uses Table 4.2's fitted coefficients to compute model-implied scenarios: the predicted percentage output changes from a hypothetical import or concentration shift, holding other factors constant. This is described as an Exploratory Model-Implied Scenario Tool, not a causal policy simulator, and every result should be read as an illustrative association, not a causal forecast. Genuine causal estimation would require exogenous variation, natural experiments, shift-share instruments, event-study or difference-in-differences designs, or a structural CGE model none of which this exercise provides.

For a subsector with import elasticity 0.1998 (illustrative despite not being statistically decisive under bootstrap inference), a hypothetical X percent import reduction implies a model-implied output change of $(1+X)^{0.1998} - 1$.

Table 4.12. Illustrative Model-Implied Output Association with a Hypothetical Import Disruption, by Subsector and Shock Magnitude (2025 Baseline): Exploratory and Non-Causal

Subsector	2025 Baseline Output (\$B)	–10% Imports	–20% Imports	–30% Imports
Computer and electronic products	425.6	\$416.7B (–2.08%)	\$407.0B (–4.36%)	\$396.3B (–6.88%)
Chemical products	1,016.7	\$995.5B (–2.08%)	\$972.4B (–4.36%)	\$946.8B (–6.88%)
Food and beverage and tobacco products	1,268.3	\$1,241.9B (–2.08%)	\$1,213.0B (–4.36%)	\$1,181.1B (–6.88%)
Primary metals	303.4	\$297.1B (–2.08%)	\$290.2B (–4.36%)	\$282.5B (–6.88%)

This arithmetic implies an identical percentage association for a given shock across every subsector, differing only in dollar terms; Section 4.4.4's interaction test found no significant evidence that concentrated subsectors are more sensitive, so this uniform-elasticity property is consistent with what the data shows, within this panel's statistical power. A parallel example: a

one-standard-deviation concentration improvement (–673 HHI points) implies a model-implied output change of approximately 2.37%; for a median-sized subsector (~\$231B), roughly \$5.5 billion presented here as an illustration of the calculation, not a statistically supported benefit estimate, since the underlying coefficient is not significant in the preferred specification (Section 4.4).

4.6.1 Comparative Illustration Across Alternative Interventions

The framework's design calls for comparing alternative interventions, not evaluating a single shock in isolation. Table 4.13 applies Table 4.2's coefficients to three named interventions for Computer and Electronic Products (highest import exposure): a tariff-style import reduction, a supplier-diversification action, and a combined approach.

Table 4.13. Illustrative Comparison of Alternative Interventions: Computer and Electronic Products, 2025 Baseline (\$425.6B); Exploratory and Non-Causal

Intervention	Assumption	Model-Implied Output Change	Resulting Output
A: Tariff-style import reduction	–20% imports	–4.36%	\$407.0B
B: Supplier diversification	–1 SD import-origin concentration	+2.43%	\$435.9B
C: Combined approach (A + B)	Both simultaneously	–2.04%	\$416.9B

This illustrates a trade-off the fitted model implies import reduction and diversification pull output in opposite directions, but two caveats apply. The concentration coefficient underlying Scenario B is not significant in the preferred specification, so its contribution is illustrative, not a supported claim. And this arithmetic treats the two interventions as independent and additive, when reducing concentration would plausibly change import volumes. A genuine policy-comparison capability to resolve this with statistical and causal confidence is future work (Section 5.6).

4.7 Decision-Support Interface

Addressing RQ6, a prototype decision-support dashboard (the Economic Intelligence Copilot) draws on this real panel, presenting summary indicators, output trends, the results in this chapter, and the Exploratory Model-Implied Scenario Tool, labeled consistently with this chapter's non-causal framing throughout. The interface has not been formally evaluated against alternative formats, so RQ6 is only partially answered; systematic evaluation with real policymakers or manufacturer users is a priority for future research (Chapter Five).

4.8 Correspondence with the Proposed Five-Phase Methodology

This chapter's empirical work constitutes real evidence touching the first three phases of Chapter Three's methodology but does not validate every technique in that broader architecture. Structural gravity models, full VAR systems, network-based dependency mapping, and CGE modeling are not implemented here; the methods actually estimated are a two-way fixed-effects regression and its robustness variants, a rolling-origin forecasting exercise, a computed concentration index, and arithmetic scenario calculations. Table 4.14 states plainly what has and has not been demonstrated.

Table 4.14. Correspondence Between Empirical Work Completed and the Proposed Five-Phase Methodology

Phase	Demonstrated with real data	Not yet demonstrated
Phase 1: National Trade and Industrial Intelligence Assessment	Integration of real federal data across BEA, BLS, and USITC; documented data-infrastructure gaps (Section 4.2); baseline resilience and concentration indicators (Section 4.3)	A formal dependency network linking specific foreign suppliers to specific domestic manufacturers and downstream industries
Phase 2: Predictive Economic Framework Development	A two-way fixed-effects econometric model with honestly reported inference (Sections 4.4–4.4.7); a pseudo-out-of-sample, rolling-origin forecasting exercise (Section 4.5); a real, computed import-origin concentration index; an employment-impact model, honestly reporting a null result (Section 4.4.7); a comparative illustration across named alternative interventions (Section 4.6.1); a documented human-in-the-loop validation protocol specified for pilot implementation (Section 3.9.1)	The gravity-model, VAR, and CGE components described in Section 3.8 as part of the framework's intended architecture; economic-impact models estimating disruption effects on prices specifically (only employment was estimated); the human-in-the-loop protocol operated in practice rather than only specified; a policy-comparison capability that resolves trade-offs with statistical and causal confidence rather than illustrating them arithmetically
Phase 3: Prototype Validation and Pilot Implementation	A decision-support dashboard interface drawing on the real panel (Section 4.7); an out-of-sample forecast evaluation against a naive benchmark (Section 4.5); a battery of regression diagnostics (Section 4.4.2)	Retrospective validation of model predictions against specific historical disruption episodes; formal evaluation of decision-support interface formats (Section 4.7); a standardized, documented procedure for replicating the framework across additional industries
Phases 4–5: National Scaling and Continuous Improvement	Not applicable at this stage	These phases are institutional and programmatic in nature and are outside the scope of a single empirical analysis

This statement represents an honest account of the study's progress rather than a claim that the framework is fully complete. The findings show that its main data-integration and modeling concepts are workable. However, the results also show that the pseudo-out-of-sample forecasts

were difficult to improve on the naive benchmark, the relationship between concentration and output was consistent in direction but not statistically significant, and differences in vulnerability across concentration levels were not supported. The gravity-model, network-mapping, VAR, and CGE components remain areas for future development.

4.9 Summary of Findings

This chapter presented results from 19 U.S. manufacturing subsectors covering 2015–2025, based on the six research questions in Chapter One. The findings showed gaps in data infrastructure (RQ1). Trade activity was positively associated with output, but the relationship was not significant under the preferred wild-cluster bootstrap method, while import concentration had a negative but non-significant relationship with output (RQ2, RQ4). The employment model also found no significant relationship. Forecasting results showed that the naive benchmark was difficult to outperform (RQ3). The trade $\times$ concentration interaction was not significant, while a three-policy comparison illustrated the framework's intended policy application (RQ5). A prototype dashboard was developed, but not formally evaluated, and the human-in-the-loop protocol had not yet been implemented (RQ6). Chapter Five discusses the implications, contributions, limitations, and future research.

CHAPTER FIVE: DISCUSSION, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Discussion of Findings

The findings in Chapter Four generally support the study's main idea, while also showing the need to interpret some of the results with caution. The panel regression in Section 4.4 shows a positive association between trade activity and output under asymptotic inference. However, the wild-cluster bootstrap, which is more suitable for the 19 industry clusters in the panel, does not show a statistically significant relationship. Import-origin concentration is negative across the different model specifications and significant in three of four alternatives (Table 4.3), but it is not significant in the preferred two-way fixed-effects and interaction models. Therefore, the evidence does not provide conclusive support for the concentration-risk hypothesis. Instead, it highlights the need for a larger panel to examine this relationship more reliably.

The pseudo-out-of-sample forecasting results in Section 4.5 show that the simple naive persistence model is difficult to outperform. This should not be viewed as a negative finding. Rather, it provides the honest benchmark comparison required in Section 3.9 and shows that the framework's evaluation standards are being applied rigorously instead of being presented only as methodological claims.

5.2 Contributions to the Field

This study makes four main contributions. First, it provides a clear and replicable method for combining fragmented federal and industry data (Section 2.2). Second, it proposes a framework that links supply-chain dependency mapping with econometric forecasting (Section 2.3), although the formal network component is still under development and has not yet been fully implemented or validated (Table 4.14). Third, it evaluates a hybrid econometric and machine-learning forecasting approach honestly, with the pseudo-out-of-sample results showing that it does not currently outperform the naive benchmark. Fourth, it develops a prototype approach for comparing policy scenarios and national indicators. These tools are presented as illustrative and non-causal, while also helping to address the existing gap in practical decision-support tools (Section 2.7).

Overall, these contributions demonstrate an applied framework that brings together established methods such as panel econometrics, machine-learning forecasting, concentration measurement, and decision-support tools. Although these methods are often used separately in this area, the proposed framework combines them in a single approach that can be expanded through the Phase 4 and Phase 5 methodology beyond the industries covered in this pilot study.

5.3 Policy Implications and National Significance

This section explains the need for further development and testing of the pilot framework, the economic benefits that may reasonably be expected, and its potential contribution to competitiveness and resilience. These discussions are based on the actual findings reported in Chapter Four rather than on the framework's intended design alone.

5.3.1 Policy Relevance and Case for Further Pilot Implementation

Three main findings support continued development of the pilot framework. First, the framework can be developed using data that policymakers already have access to. All variables were obtained from existing and publicly funded federal data systems, meaning that the main requirement is effective data integration rather than the collection of entirely new data. Second, the diagnostic analysis in Sections 4.4.1–4.4.2 demonstrates a strong evaluation process. The study applies a Hausman test, Mundlak test, wild-cluster bootstrap, and several model specifications, while reporting results that do not always support the main hypotheses. This provides greater confidence in the overall evaluation process than relying on a single statistical estimate. Third, the forecasting analysis in Section 4.5 shows that naive persistence is difficult to outperform. Although this limits the current forecasting gains, it also demonstrates that the evaluation procedures in Section 3.9 can identify weak or overconfident forecasting claims before they are presented to policymakers. This type of safeguard is important for a national decision-support framework.

In addition to these findings, the federal systems examined in this study, including those of the BEA, BLS, and USITC, do not provide a single, regularly updated panel that combines output, employment, productivity, and trade-concentration data. These datasets are produced separately by the different agencies and therefore require manual reconciliation (Section 4.2). This finding is based on the systems reviewed in this study and should not be taken as a complete assessment of all federal data systems. Creating an integrated system, even at the pilot's 19-subsector level, could

reduce the repeated effort required to bring these datasets together for analysis. The need for such a system is also relevant to current policy priorities. The Trade Policy Agenda identifies the protection of supply chains for critical minerals and other strategic sectors as an important national priority (Office of the United States Trade Representative, 2026). This provides a clear policy context for the framework and demonstrates its relevance to an existing national policy objective.

5.3.2 Potential Economic Relevance and Analytical Efficiency

The type of estimate that the framework could produce in future is illustrated by the scenario tool presented in Section 4.6. A detailed dollar-value example is provided there rather than in this section because the concentration coefficient is not statistically significant in the preferred model. Therefore, it would not be appropriate to present it as an expected economic benefit. A reliable, decision-ready estimate will require further research using a larger panel, as discussed in Section 5.6.

A more certain benefit of the framework is improved analytical efficiency. The data-integration process described in Section 4.2 can be reused once it has been established. Although compiling the 209 observations required considerable initial effort, an institutionalized system would avoid repeating the same process for every new policy question. Adding new year's or subsectors under Phase 4 (Section 3.5) would also require less effort than the initial development. This creates a system in which most of the cost occurs during the initial setup, while the additional cost of maintaining and expanding it remains relatively low, as is common with effective federal data infrastructure.

5.3.3 Impact on Manufacturing Competitiveness and Supply-Chain Resilience

The panel data in Appendix A already provides useful information for identifying potential risks before any policy action is taken. For example, the 2025 data show that Wood Products (HHI 1,962) and Printing (HHI 1,736) have the highest levels of import concentration, while Food and Beverage/Tobacco (HHI 837) have the most diversified import sources. This provides a useful picture of differences in exposure across subsectors and can help identify areas that may require closer monitoring, regardless of whether the concentration-output relationship is statistically significant.

Regarding competitiveness, both imports and exports have positive coefficients in Table 4.2. However, neither relationship is statistically significant under the preferred bootstrap method (Section 4.4.2). The results therefore provide only a possible indication that subsectors with greater involvement in international trade may also have higher productivity. This should be treated as a hypothesis for future research rather than a confirmed finding. Competitiveness policies should therefore consider the potential benefits of import integration alongside export promotion rather than assuming that imports are always harmful.

Regarding resilience, the trade $\times$ concentration interaction test in Section 4.4.4 found no statistically significant evidence that highly concentrated subsectors are more sensitive to shocks. This is a genuine null finding and is reported directly rather than being excluded from the results. The finding suggests that policymakers should be cautious when using trade-risk tools that assume all subsectors respond to shocks in the same way. This study can make this caution more confidently because the interaction between trade activity and concentration was tested directly rather than simply identified as a possible limitation.

5.4 Recommendations for Implementation

Four recommendations arise from the findings and development of this study. First, data integration should be prioritized before predictive modeling, since the quality of forecasts and scenario analysis depends largely on the quality of the underlying data (Section 3.2). Second, improving visibility beyond lower supply-chain tiers should remain an ongoing priority, because the current country-level concentration measure cannot capture firm-level and multi-tier dependencies. Third, human-in-the-loop validation should be incorporated as a continuing governance measure rather than treated as a one-time validation stage, because forecasting always involves some level of uncertainty (Sections 3.9.1 and 3.10). Fourth, implementation should be carried out gradually on an industry-by-industry basis, using the modular components developed in Phase 3 instead of attempting immediate national deployment. This approach would allow the framework to be tested and improved at each stage.

5.5 Limitations of the Study

This study has several limitations. The results presented in Chapter Four are based on real federal data and are not simulated. However, they come from a relatively small pilot panel covering 19

manufacturing subsectors over 11 years. Therefore, the findings should not be treated as a complete national assessment, causal estimates, or results that can be widely generalized beyond the study panel. A larger panel would be needed for national-level application, while causal conclusions would require a research design capable of identifying exogenous sources of variation.

The forecasting models also have the common limitations of econometric and machine-learning methods, including lower accuracy when unexpected structural changes occur and sensitivity to the quality of the available data (Section 3.10). In addition, the current concentration measure does not provide adequate information about lower-tier supply-chain relationships. Finally, although the 19-subsector pilot covers different types of supply-chain dependencies, it does not represent the full range of U.S. manufacturing. Further expansion through Phase 4 (Section 3.5) would therefore be necessary to assess whether the framework performs effectively across a wider range of industries.

5.6 Recommendations for Future Research

Future research should focus on implementing the framework more fully using a larger number of industries and carrying out systematic retrospective validation across a wider range of historical disruption events. The correctly signed but non-significant concentration result (Section 4.4), the non-significant interaction in Section 4.4.4, and the difficulty of improving on naive persistence in Section 4.5 all point to the need for a larger panel. A broader dataset would provide greater statistical power to test relationships that are theoretically expected but remain inconclusive on the current scale.

Further studies should also develop the formal supply-chain dependency network proposed in Section 2.3 and Table 4.14. The fixed-effects model could be extended to more detailed price and employment-impact models, while dynamic panel GMM methods (Section 3.8.1) could be examined to address the serial correlation identified in Section 4.4.2. Developing a real-value version of the panel using subsector-specific price indices should also be prioritized because the current analysis relies on nominal dollar values (Section 4.3). Finally, future research should examine governance arrangements for maintaining Phase 5's continuous updating process, including effective data-sharing arrangements among federal agencies, universities, and industry.

Such arrangements would be important for sustaining the framework and supporting its eventual national expansion.

5.7 Conclusion

This study addressed an identified gap in U.S. trade and industrial-policy analysis, particularly the limited availability of integrated and forward-looking tools that combine fragmented federal data with supply-chain analysis and predictive methods. Based on the literature reviewed in Chapter Two, Chapter Three developed a five-phase methodology covering data integration, supply-chain dependency mapping, hybrid forecasting, and scenario analysis. The framework was then tested at a pilot level in Chapter Four using real federal data.

The resulting Predictive Trade and Industrial Intelligence Framework provides a replicable approach for improving the evidence base for U.S. trade policy, manufacturing resilience, and industrial competitiveness. Although full national implementation would require further development beyond the scope of this study, the framework and its pilot-scale empirical evaluation provide a structured and practical starting point for future research and wider application.

5.8 Data and Code Availability

The panel dataset, selected analysis scripts, documented methodology, and the decision-support dashboard prototype, the Economic Intelligence Copilot, are available through an open-source repository under the MIT License at <https://github.com/aolusoji/economic-intelligence-copilot>. The live Economic Intelligence Copilot decision-support interface is available at <https://aolusoji.github.io/economic-intelligence-copilot/>. The repository contains materials supporting key components of the empirical analysis, including data-construction procedures, selected econometric and forecasting workflows, and documentation describing the structure of the dataset and analytical framework. These resources are intended to support transparency, reproducibility of key aspects of the study, and further development of the framework. The live interface includes the exploratory scenario-analysis functionality discussed in Section 4.6 and is deployed from the same repository. Additional analytical materials and extensions may be added as the framework continues to develop.

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APPENDIX A: REAL PANEL DATASET

A.1 Dataset Description and Summary Statistics

This appendix presents descriptive statistics and a full-year snapshot of the real panel dataset underlying Sections 4.2–4.6. The complete dataset (209 industry-year observations, 2015–2025) is in the accompanying workbook, *Manufacturing_Panel_Dataset.xlsx*, with a full data-sources and construction-methodology sheet.

Table A.1 reports summary statistics across all 209 observations.

Table A.1. Descriptive Statistics, Full Panel ($N = 209$; 19 NAICS-3 Manufacturing Subsectors $\times$ 11 Years, 2015–2025)

Statistic	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI (0–10,000)
Mean	330.1	658.1	76.0	124.6	97.9	1,558.4
Std. Deviation	287.5	462.3	72.3	115.0	6.0	672.9
Minimum	15.7	79.0	4.4	4.9	78.6	645.9
25th Percentile	124.3	368.6	11.8	36.9	93.8	987.7
Median	231.4	427.4	59.4	94.9	99.3	1,481.8
75th Percentile	412.6	944.2	118.5	168.5	101.2	1,907.6
Maximum	1,268.3	2,098.7	302.7	709.4	115.0	3,605.9

A.2 Full-Panel Snapshot: 2025

Table A.2 presents the panel's most recent year (2025) for all 19 subsectors, illustrating the cross-sectional variation the panel regression exploits. Full annual detail for 2015–2025 is in the accompanying workbook.

Table A.2. Manufacturing Subsector Panel Data, 2025

Subsector	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI
Apparel and leather and allied products	22.6	79.0	9.38	121.62	115.0	1,065
Chemical products	1,016.7	900.5	293.20	411.28	90.0	971
Computer and electronic products	425.6	997.7	302.68	709.35	114.5	1,299
Electrical equipment, appliances, and components	199.3	429.0	76.01	192.34	93.4	1,243

Subsector	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI
Fabricated metal products	464.7	1,436.2	60.74	88.11	89.2	946
Food and beverage and tobacco products	1,268.3	2,098.7	92.77	149.06	78.6	837
Furniture and related products	85.3	337.9	4.81	39.90	89.7	1,643
Machinery	437.3	1,089.1	160.21	233.46	93.7	762
Miscellaneous manufacturing	181.9	612.7	84.06	132.91	82.8	955
Motor vehicles, bodies and trailers, and parts	750.7	967.4	175.75	246.72	100.0	1,794
Nonmetallic mineral products	174.3	417.0	11.40	27.47	88.2	892
Other transportation equipment	409.2	776.1	141.00	197.94	100.0	1,794
Paper products	221.4	355.0	23.82	25.25	87.4	1,584
Petroleum and coal products	617.6	109.7	117.88	61.22	97.4	830
Plastics and rubber products	297.4	704.0	40.24	74.86	90.0	944
Primary metals	303.4	363.9	154.95	210.51	82.7	1,150
Printing and related support activities	85.1	343.8	4.91	5.84	93.5	1,736
Textile mills and textile product mills	47.2	175.2	10.82	29.40	100.9	1,505
Wood products	172.9	403.9	7.90	23.55	93.6	1,962

Food and beverage/tobacco is the largest subsector by output and employment, with the lowest import-origin concentration (HHI = 837), consistent with a diversified agricultural input base. Wood Products and Printing show the highest concentration (above 1,700). Computer and electronic products show the largest trade imbalance in absolute terms (\$302.7B exports vs. \$709.3B imports), consistent with documented U.S. dependence on foreign semiconductor and electronics manufacturing capacity (Chapter One).